

SOOYONG KIM

Robotics & Autonomous Systems Engineer

sooyongtech.dev | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Robotics and autonomous systems engineer with a strong mechanical-engineering research foundation in dynamics, mathematical modeling, state estimation, sensing, control, simulation, and experimental validation. Experienced in ROS/ROS 2, mobile robotics, sensor fusion, perception, path planning, and model-based robotics, with research experience in autonomous mobile robots at the University of Waterloo and vibration-based system identification and model validation at the University of Victoria. Current work focuses on analytical and simulation-based robotics development using Python, C++, ROS 2, and Isaac Sim, with emphasis on kinematics, dynamics, control, perception, and quantitative cross-validation. Strong focus on reconciling mathematical models, simulation behavior, and quantitative results to identify discrepancies in modeling assumptions, coordinate frames, interfaces, and implementation.

EDUCATION

University of Waterloo PhD Program in Mechanical & Mechatronics Engineering Voluntary Withdrawal	2020–2021
University of Victoria Master of Applied Science (MASC), Mechanical Engineering Thesis: Predicting Regenerative Chatter in Turning Using Operational Modal Analysis	2016–2019
Pusan National University Bachelor of Science (B.Sc.), Mechanical Engineering Undergraduate Thesis: Investigation of SOFC Injecting Natural Gas Directly	2005–2012

SELECTED ROBOTICS PROJECTS

UR5e Modeling, Dynamics & Control

Personal Robotics Project

Python · C++ · ROS 2 · Isaac Sim · Rigid-Body Dynamics · Control

- Reconstructed and validated a UR5e robot model from URDF-defined kinematic, inertial, and frame parameters.
- Developed analytical kinematics and dynamics models and cross-validated robot poses, geometric Jacobians, joint-space mass matrices, and gravity terms against Isaac Sim.
- Reconciled coordinate frames, joint conventions, base orientation, fixed-base conditions, and simulator configuration to ensure physically consistent comparison between analytical and simulated models.
- Implemented quantitative error analysis to identify discrepancies between mathematical predictions and simulator outputs rather than treating simulation results as ground truth.
- Developing ROS 2 and C++ implementations for manipulator control, trajectory tracking, and model-based control using the validated robot model.
- Extending the platform toward depth-camera-based perception and manipulation experiments.

ROS 2 EKF Sensor Fusion

Personal Robotics Project

ROS 2 Humble · Gazebo

- Implemented an Extended Kalman Filter pipeline for mobile-robot state estimation using wheel odometry and IMU measurements.
- Integrated simulated sensor streams, coordinate frames, and estimated robot states within a Gazebo-based ROS 2 environment.
- Compared raw wheel-odometry trajectories against EKF-fused state estimates and ground-truth motion.
- Quantified localization error under curved and extended trajectories to evaluate estimator behavior and sensor-drift correction.
- Documented the implementation and validation process as a reproducible robotics experiment.

Real-Time Vision Perception Pipeline

Personal Robotics Project

ROS 2 · PyTorch · Ultralytics YOLO · OpenCV · DeepSORT

- Developed a ROS 2 perception pipeline integrating YOLO-based instance segmentation, multi-object tracking, image processing, and quantitative performance logging.
- Achieved 31.95 FPS across 2,136 steady-state frames, with 18.26 ms average processing latency and 24.87 ms P95 latency on an NVIDIA RTX 5060 Laptop GPU.
- Measured 8.70 ms average inference latency and 12.52 ms P95 inference latency, processing scenes containing up to 11 detected objects per frame.
- Structured the implementation as reusable ROS 2 nodes with launch configuration, CSV-based performance reporting, Git version control, and documented experimental results.

LiDAR-Camera 3D Perception & Fusion

Personal Robotics Project

ROS 2 · Isaac Sim · LiDAR · RGB Camera · OpenCV · 3D Geometry

- Developed a LiDAR-camera fusion pipeline using synchronized simulated sensor streams in Isaac Sim.
- Calibrated camera intrinsics and LiDAR-to-camera extrinsics and implemented 3D-to-2D projection between sensor coordinate frames.
- Combined image-based object detection and segmentation with projected LiDAR measurements for object-level distance estimation.
- Investigated geometric consistency, sensor-frame alignment, projection accuracy, and depth robustness in warehouse scenes.
- Documented the sensor model, coordinate transforms, validation procedure, and fusion results for reproducible evaluation.

RESEARCH EXPERIENCE

University of Waterloo

May 2020 – Dec 2021

Research Assistant, Mechanical & Mechatronics Engineering

Waterloo Mechanical Systems and Control Lab

- Conducted mobile robotics research involving autonomous navigation, path planning, localization, sensing, and physical-robot experimentation.
- Investigated transfer-learning approaches for improving learning efficiency in robot semantic tasks.
- Developed LiDAR-camera fusion methods for semantic road segmentation using deep neural networks and the KITTI dataset.
- Implemented model-predictive-control-based mobile-robot navigation and conducted physical experiments to evaluate path-following performance.
- Supported research on path-planning optimization for indoor mobile robots operating in dynamic environments.
- Conducted physical experiments using Summit mobile robots, motion-capture systems, sensors, and associated robotic hardware.
- Supported physical validation of motion-aware navigation and planning approaches.
- Maintained robotic systems, sensors, accessories, and experimental platforms used for laboratory research.
- Developed robot-operation documentation and technical instructional material.

University of Victoria

Aug 2017 – Apr 2019

Research Assistant, Dynamics and Digital Manufacturing Lab

Operational Modal Analysis for Chatter Prediction in Turning

- Developed an Operational Modal Analysis approach for estimating machining stability from in-process vibration measurements before regenerative chatter fully developed.
- Modeled turning dynamics using multi-degree-of-freedom, state-space, and time-delay formulations.
- Performed numerical simulation, pole identification, modal damping estimation, and stability analysis in MATLAB.
- Designed and conducted instrumented machining experiments using accelerometers, National Instruments data-acquisition hardware, acoustic measurements, and experimental modal analysis.
- Applied correlation analysis, power spectral density analysis, Time Domain Poly Reference identification, and modified Least Squares Complex Exponential methods.
- Investigated separation of structural modes from deterministic spindle harmonics in operational vibration measurements.
- Conducted finite-element and modal analyses related to machining-system structural dynamics.
- Validated analytical and numerical stability predictions against physical machining experiments.

Micromilling Dynamics & Laser Doppler Vibrometry

- Conducted non-contact vibration measurements of a high-speed micromilling system using a Laser Doppler Vibrometer.
- Used broadband acoustic excitation to characterize tool-tip structural dynamics.
- Applied Operational Modal Analysis to estimate modal frequencies and damping ratios from output-only vibration measurements.
- Compared OMA results with conventional impact-hammer Experimental Modal Analysis.
- Investigated spindle-harmonic interference at multiple rotational speeds and applied modified LSCE methods to distinguish harmonic excitation from structural modes.
- Performed MATLAB-based signal processing using correlation functions, power spectral density, modal identification, and stabilization analysis.

University of Victoria

Jun 2016 – Jul 2017

Research Assistant, Laboratory of Advanced Multi-scale Manufacturing

- Fabricated and tested fiber-optic sensors, including Long Period Grating (LPG) and Fiber Bragg Grating (FBG) devices.
- Conducted femtosecond-laser machining of glass and other materials.
- Fabricated microfluidic channels in glass using femtosecond-laser processing.
- Investigated glass welding for microfluidic devices without intermediate bonding layers.
- Conducted nanoparticle coating experiments on glass substrates.
- Supported experimental setup, fabrication, testing, and engineering analysis.

PROFESSIONAL EXPERIENCE

MKS Investments

2023–2026

Manager

- Reviewed and validated technical and market analyses produced by internal analysts and AI-assisted workflows, identifying incorrect assumptions, data inconsistencies, unsupported claims, and source-quality issues.
- Cross-checked technical feasibility, benchmarks, source data, and market evidence to improve the reliability of technology-development and investment decisions.
- Managed technology and market research across emerging industries, synthesizing technical, commercial, and competitive information.
- Evaluated technologies and translated technical findings into decision-oriented recommendations for management and external stakeholders.
- Supported business-development activities through technology evaluation, research synthesis, and market analysis.

Kyungdong LNG

2010

Co-op Student

- Monitored operating conditions of LNG storage tanks and LNG pressure-reduction facilities.
- Supported routine operational monitoring and equipment-status checks.

TEACHING EXPERIENCE

University of Waterloo

Autonomous Mobile Robotics — MTE 544

Teaching Assistant | Fall 2021

- Customized laboratory manuals for students in the Mechatronics Engineering program.
- Provided instruction and technical support for programming in the Robot Operating System (ROS) environment.
- Supported laboratory exercises involving ROS, SLAM, localization, navigation, and mobile robotics.
- Evaluated assignments and examinations and provided technical feedback.

University of Waterloo

Autonomous Mobile Robotics — ME 597c

Teaching Assistant | Winter 2021

- Developed and modernized laboratory manuals for remote robotics instruction.
- Supervised laboratory activities and evaluated laboratory reports.
- Provided instruction and troubleshooting support for ROS-based mobile robotics.
- Created demonstration videos illustrating mobile-robot maneuvering and navigation behavior.

Robotics Laboratory Development

- Prepared and validated ROS, Gazebo, MATLAB, and TurtleBot environments for remote laboratory exercises.
- Developed and validated laboratory material covering ROS fundamentals, SLAM, AMCL localization, and autonomous navigation.
- Developed AMCL parameter-study exercises examining particle count, measurement noise, resampling behavior, and computational trade-offs.
- Developed sampling-based path-planning exercises involving RRT and RRT*, robot geometry, obstacle inflation, collision validation, Reeds-Shepp and Dubins state spaces, and Pure Pursuit tracking.
- Supported live instruction, debugging, experimental verification, grading, and laboratory-report assessment.

University of Victoria

Fall 2017

Introduction to Mechanical Vibrations — MECH 330

Teaching Assistant

- Supported students with mechanical-vibration theory, analytical problem solving, and course material.
- Evaluated assignments, quizzes, and tests.
- Provided academic feedback and supervised examinations.

University of Victoria

Spring 2018

Mechanics of Solids II — MECH 320

Teaching Assistant

- Managed and supervised laboratory activities.
- Guided students through experimental procedures and engineering analysis.
- Maintained laboratory safety and proper experimental practice.
- Evaluated laboratory reports and assignments.

University of Victoria

Summer 2018

CAD-Aided Manufacturing — MECH 460

Teaching Assistant

- Managed laboratory operations and scheduling.
- Updated laboratory manuals and manufacturing-workflow documentation.
- Evaluated student project designs and supervised CNC machining activities.
- Provided instruction and guidance in CAD/CAM software and manufacturing workflows.
- Reviewed student designs, toolpaths, machining programs, coordinate systems, and setup details before physical machining.
- Supported SolidWorks, MasterCAM, VERICUT, G-code verification, and Haas 3-axis CNC machining.
- Assisted students from CAD design through CAM programming, machining verification, and final part production.
- Evaluated laboratory reports and provided technical feedback.

UNDERGRADUATE RESEARCH & ENGINEERING

Solid Oxide Fuel Cell System Fueled Directly by Natural Gas

2011–2012

Pusan National University

- Developed and evaluated a Solid Oxide Fuel Cell (SOFC) system designed to utilize natural gas directly without an external reformer.
- Conducted fuel-cell system setup, experimental testing, and performance analysis.
- Investigated direct natural-gas utilization in tubular-type solid oxide fuel cells.
- Developed the research into a Korean patent application.
- Received university and external awards associated with the project.

Undergraduate Thesis: Investigation of SOFC Injecting Natural Gas Directly

SELECTED PUBLICATION

S. Kim and K. Ahmadi, "Estimation of vibration stability in turning using operational modal analysis," Mechanical Systems and Signal Processing, 2019.

PATENT

Solid Oxide Fuel Cell System Fueled By Natural Gas Directly
Korean Patent Application No. 10-2012-0109237
Inventor: Sooyong Kim et al.

2012

SCHOLARSHIPS & AWARDS

- University of Waterloo Graduate Funding — CAD 40,000 | 2020-2021
- University of Victoria Scholarship — CAD 27,000 | 2016-2019
- Pusan National University Scholarship — CAD 10,000 | 2008-2011
- Grand Prize, Integrated Capstone Design Competition | 2011
- Grand Prize, Research Assessment of Industry-University Collaboration | 2011
- Excellence Award, Engineering Students Technology & Science Competition | 2011

TEACHING CERTIFICATIONS

Certificate in University Teaching
Center for Teaching Excellence, University of Waterloo
Certificate in TA Training
Division of Learning and Teaching Support and Innovation, University of Victoria

Sep 2020

Sep 2017

CORE TECHNOLOGIES

Robotics & Control: ROS 2, ROS, Nav2, SLAM, AMCL, Localization, Path Planning, RRT/RRT*, Pure Pursuit, Model Predictive Control, Manipulator Kinematics, Rigid-Body Dynamics

State Estimation & Sensor Fusion: EKF, IMU, Wheel Odometry, LiDAR-Camera Fusion, Coordinate Transformations, Sensor Calibration

Perception: OpenCV, YOLO, Object Tracking, Instance Segmentation, Semantic Segmentation, RGB-D Perception

Simulation & Robotics Platforms: Isaac Sim, Gazebo, TurtleBot, Summit XL, UR5e

Dynamics & System Analysis: State-Space Modeling, Time-Delay Systems, Numerical Analysis, System Identification, Operational Modal Analysis, Experimental Modal Analysis, Stability Analysis, Signal Processing, PSD, Correlation Analysis, FEA

Programming & Computing: Python, C++, MATLAB, Linux, Git, GitHub, Jupyter